

Differentiating Implicit Relations

The idea, and the bookkeeping model. In a relation such as $x^2 + y^2 = 25$ the variable y is not written as a formula in x , but on any piece of the curve where y is a differentiable function of x we may still differentiate both sides with respect to x . Every y carries a factor $\frac{dy}{dx}$ by the chain rule. Differentiate term by term:

Term	Derivative with respect to x	Rule used
x^3	$3x^2$	power rule
y^3	$3y^2 \frac{dy}{dx}$	power rule + chain rule
xy	$y + x \frac{dy}{dx}$	product rule
x^2y	$2xy + x^2 \frac{dy}{dx}$	product rule + chain rule
a constant	0	—

- After differentiating, collect every term containing $\frac{dy}{dx}$ on one side, factor it out, and divide. The answer normally involves *both* x and y — that is expected, not a mistake.
- A slope at a point needs both coordinates. Before using a point, check that it actually lies on the curve by substituting it into the original relation.
- The tangent is horizontal where the numerator of $\frac{dy}{dx}$ is zero (with the denominator nonzero), and vertical where the denominator is zero (with the numerator nonzero).

1. Differentiate both sides of

$$x^2 + y^2 = 25$$

with respect to x and solve for $\frac{dy}{dx}$.

Then, on the upper half of this circle, $y = \sqrt{25 - x^2}$ is an explicit function. Substitute that expression for y into your implicit answer and confirm it agrees with the derivative you would get by differentiating $\sqrt{25 - x^2}$ directly. Write both forms.

Answer: _____

2. For the relation

$$xy = 12,$$

find $\frac{dy}{dx}$. Verify that $(3, 4)$ lies on the curve, then find the slope of the curve there.

Answer: _____

3. For the relation

$$x^2 + xy + y^2 = 7,$$

find $\frac{dy}{dx}$ in terms of x and y . Verify that $(1, 2)$ lies on the curve, then evaluate the slope there.

Answer: _____

4. The *folium of Descartes* is the curve

$$x^3 + y^3 = 9xy.$$

Find $\frac{dy}{dx}$, verify that $(2, 4)$ lies on the curve, and find the slope of the curve at that point.

Answer: _____

5. For the relation

$$x^2y + y^3 = 10,$$

find $\frac{dy}{dx}$. Verify that $(1, 2)$ lies on the curve, then evaluate the slope there. Note where the product rule and the chain rule each had to be used.

Answer: _____

6. The ellipse

$$x^2 + 4y^2 = 25$$

passes through $(3, 2)$. Verify that it does, find the slope of the curve at that point, and write an equation of the tangent line there in point-slope form.

Answer: _____

7. Horizontal tangents. For the curve

$$x^2 + xy + y^2 = 12,$$

find $\frac{dy}{dx}$, then find *all* points on the curve where the tangent line is horizontal.

Set the numerator of $\frac{dy}{dx}$ equal to zero to get a relation between x and y , substitute that into the original equation to get an equation in x alone, and solve. Give both points, and confirm each satisfies the original relation.

Answer: _____

8. Challenge — second derivative. For

$$x^2 + y^2 = 25,$$

you found $\frac{dy}{dx} = -\frac{x}{y}$. Differentiate that a second time with respect to x , substituting your expression for $\frac{dy}{dx}$ wherever it appears, and simplify using the original relation to write $\frac{d^2y}{dx^2}$ in terms of y alone.

Then write the same second derivative in terms of x by substituting $y = \sqrt{25 - x^2}$, and say what the sign of the result tells you about the concavity of the upper semicircle.

Answer: _____